

Related Work Landscape for Angular Manifold Routing

This section summarizes the closest prior research directions related to Angular Manifold Routing. The goal is to position the contribution accurately with respect to existing literature on geometric representation learning, mixture-of-experts routing, and sparse compute architectures. The central empirical observation motivating this work is that routing tokens by angular sectors on a normalized hypersphere appears to produce sublinear routing occupancy. Empirically: $E(K) \approx c * K^\alpha$, where $\alpha < 1$. This suggests that routing sparsity can emerge from geometric structure rather than from learned gating networks. The literature below represents the closest conceptual neighbors to this idea.

Hyperbolic and Non-Euclidean Representation Learning

Research on hyperbolic neural networks and other non-Euclidean representation spaces explores how curved geometries can represent hierarchical data efficiently [1][2]. Hyperbolic embeddings are particularly effective for representing taxonomies, knowledge graphs, and hierarchical language relationships. Closest overlap: recognition that representation geometry affects model efficiency. Key difference: these works focus on representation structure rather than routing mechanisms. They do not introduce deterministic angular sector routing or analyze routing occupancy scaling.

Standard Mixture-of-Experts Routing

Modern mixture-of-experts architectures rely on learned gating networks that score experts and select a subset for each token [3][4]. These systems enable conditional computation but require training a routing function and typically include load-balancing losses. Closest overlap: sparse compute through expert routing. Key difference: routing is learned rather than derived from geometric partitioning of embedding space.

Hyperspherical Embedding Methods

Several works explore hyperspherical embeddings in which token representations are normalized to lie on a sphere and compared using cosine similarity [5]. These approaches leverage angular relationships between vectors. Closest overlap: use of hyperspherical geometry. Key difference: geometry is used for similarity scoring rather than as a deterministic routing partition.

Spherical or Geometric Routing Variants

Some domain-specific systems introduce spherical routing ideas or geometric similarity mechanisms for selecting experts or attention regions [6]. Closest overlap: explicit spherical routing language. Key difference: these systems still rely on learned similarity scoring rather than fixed angular sector discretization.

Geometry-Aware Aggregation

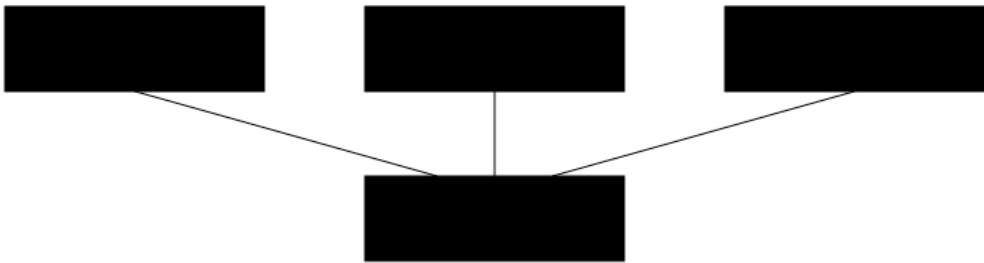
Other work investigates preserving geometric relationships when combining outputs from multiple experts, sometimes decomposing vectors into radial and angular components on manifolds [7]. Closest overlap: explicit angular and radial decomposition. Key difference: these methods affect aggregation rather than routing decisions.

Latent Routing Space Methods

Recent work attempts to improve MoE routing stability by learning improved routing spaces or low rank latent representations [8]. Closest overlap: routing quality depends on embedding geometry. Key difference: routing remains a learned transformation rather than a deterministic geometric partition.

Area of Prior Work	Main Idea	Difference From Angular Manifold Routing
Hyperbolic Representation Learning	Curved spaces encode hierarchy efficiently	Focuses on representation geometry rather than routing architecture
Standard MoE Routing	Learned gating networks select experts	Routing learned rather than determined by geometric partition
Hyperspherical Embeddings	Normalized vectors compared using cosine similarity	Geometry used for scoring rather than sector routing
Spherical Routing Variants	Similarity based routing in specific domains	Still relies on learned scoring
Geometry-Aware Aggregation	Preserves angular and radial structure in outputs	Affects aggregation not routing
Latent Routing Spaces	Learns improved routing geometry	Routing still controlled by learned function

Prior-Art Landscape Diagram



References (placeholders) [1] Hyperbolic Neural Networks [2] Poincare Embeddings for Hierarchical Representation Learning [3] Switch Transformers [4] Mixture-of-Experts Architectures [5] Hyperspherical Representation Learning [6] Spherical Expert Routing Variants [7] Geometry-Preserving Expert Aggregation [8] Latent Routing Space Methods